

# MediCore Smart General Hospital

## Project Management Practice Exercise — Progress Summary

This document summarizes the project management exercise developed for internship preparation: mastering MS Project / Primavera P6 workflows through the full lifecycle of a fictional smart-hospital construction project, from Business Case through Cost Baseline.

### 1. Project Overview (from Project Charter)

- Project: Construction of a 200-bed, ~500,000 sq ft smart general hospital
- Sponsor: MediCore Health Group Board
- Approved capital budget (Order-of-Magnitude, Charter): \$300,000,000 USD (-25%/+75%)
- Schedule constraint per Charter: substantial completion within 42 months of Notice to Proceed
- Business Case selection: NPV analysis favored the smart-technology investment option over incremental renovation

### 2. Organizational Breakdown Structure (OBS)

Top node: Enterprise, with eight functional managers directly beneath it (no separate Project Manager node exists in the OBS as built):

- Design Manager — Lead Architect, Structural Engineer, MEP Engineer, Medical Planning Specialist
- Construction Manager — Site Superintendent, Structural/MEP/Finishing Subcontractors, Site Safety Coordinator
- Smart-Systems / IT Integration Manager — EMR Integration Lead, IoT/BMS Lead, Network & Cybersecurity Specialist
- Procurement Manager — Buyer, Contract Administrator, Vendor/Supplier Coordinator
- QA/QC Manager — Quality Inspectors, Materials Testing Lab Officer
- HSE Manager — Safety Officers, Environmental Compliance Officer
- Commissioning & Accreditation Manager — Commissioning Engineer, Regulatory/Accreditation Liaison
- Financial Controller — Cost Controller, Accounts Payable Officer

### 3. Work Breakdown Structure (WBS) — Responsible Manager Mapping

| WBS    | Name                                 | Responsible Manager                           |
|--------|--------------------------------------|-----------------------------------------------|
| HCB    | Hospital construction building       | Enterprise                                    |
| HCB.1  | Project Management                   | Enterprise                                    |
| HCB.2  | Design                               | Design Manager                                |
| HCB.3  | Permits & Regulatory Approval        | Commissioning & Accreditation Manager         |
| HCB.4  | Site Preparation & Civil Works       | Construction Manager                          |
| HCB.5  | Structural Construction              | Construction Manager                          |
| HCB.6  | MEP Installation                     | Construction Manager                          |
| HCB.7  | Smart-Systems Integration            | Smart-Systems / IT Integration Manager        |
| HCB.8  | Interior Finishing & Medical Fit-out | Construction Manager (mixed)                  |
| HCB.9  | Commissioning & Testing              | Commissioning & Accreditation Manager (mixed) |
| HCB.10 | Accreditation & Handover             | Commissioning & Accreditation Manager (mixed) |

### 4. Schedule Summary

43 execution activities (A1000–A1450) were defined, sequenced with FS/SS/FF relationships, and scheduled against a US 5-day calendar with federal holiday exceptions (2025–2026). After resolving open-end activities and redundant logic, the schedule was calculated (F9).

- Project start: January 2, 2025

- Project finish: September 25, 2026
- Total duration: 435 working days (~21 calendar months)

## Critical Path

| Activity ID | Activity Name                          | Finish Date |
|-------------|----------------------------------------|-------------|
| A1000       | Develop schedule baseline              | 08-Jan-25   |
| A1080       | Conceptual design                      | 06-Feb-25   |
| A1090       | Schematic design                       | 14-Mar-25   |
| A1100       | Detailed design                        | 25-Apr-25   |
| A1160       | Submit building permit application     | 09-May-25   |
| A1170       | Building permit approval (wait)        | 24-Jun-25   |
| A1190       | Site clearing & grading                | 16-Jul-25   |
| A1200       | Foundation excavation                  | 30-Jul-25   |
| A1210       | Foundation concrete pour               | 20-Aug-25   |
| A1260       | Superstructure – Inpatient floors      | 17-Nov-25   |
| A1270       | Building envelope – facade             | 31-Dec-25   |
| A1300       | Electrical installation                | 09-Mar-26   |
| A1360       | Network & cybersecurity infrastructure | 13-Apr-26   |
| A1330       | IoT patient monitoring installation    | 26-May-26   |
| A1350       | EMR system integration                 | 23-Jul-26   |
| A1410       | Smart-systems testing & validation     | 13-Aug-26   |
| A1440       | Staff training                         | 11-Sep-26   |
| A1450       | Facility handover & closeout           | 25-Sep-26   |

*Key insight: the critical path runs primarily through design → permitting → foundation → the longest structural wing → electrical → the smart-systems integration chain — not through the specialty design disciplines (Structural Design and MEP Design both carry large float).*

## 5. Resource & Cost Summary

Resources were defined in three categories — Labor and Nonlabor (rate per day) and Material (Lump Sum, Price/Unit) — and assigned to activities. Material costs required fractional Budgeted Units (share of 1.0) rather than day-based units.

| WBS Branch                           | Budgeted Total Cost |
|--------------------------------------|---------------------|
| Project Management                   | \$1,597,500         |
| Design                               | \$457,500           |
| Permits & Regulatory Approval        | \$85,500            |
| Site Preparation & Civil Works       | \$6,251,500         |
| Structural Construction              | \$91,301,000        |
| MEP Installation                     | \$61,046,000        |
| Smart-Systems Integration            | \$23,125,000        |
| Interior Finishing & Medical Fit-out | \$39,188,000        |
| Commissioning & Testing              | \$147,000           |
| Accreditation & Handover             | \$60,000            |
| TOTAL                                | \$223,259,000       |

## 6. Open Item — Cost Baseline vs. Charter Budget

The bottom-up estimate (\$223.3M) is meaningfully below the Charter's Order-of-Magnitude authorization (\$300M), landing near the low end of the Charter's stated -25%/+75% accuracy range. Rather than artificially forcing the numbers to \$300M with a fabricated contingency line, this gap has been intentionally left open as a governance decision point to be resolved at the next stage (Sponsor review: re-baseline down, or formally allocate the ~\$77M difference as Management Reserve outside the Cost Baseline).

## 7. Next Steps

- Resolve the Cost Baseline vs. Charter Budget variance (Sponsor decision point)
- Build the time-phased S-Curve (Planned Value distribution)

- Develop the Risk Register in detail and link risks to WBS packages
- Prepare a mock EVM tracking exercise (PV/EV/AC, CPI/SPI) once a baseline is set